

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

_____show ip route ospf_____

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? _____Serial0/0/0_____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? _____65_____

Does R2 show up as an OSPF neighbor on R1? _____Yes_____

Does R2 show up as an OSPF neighbor on R3? _____No_____

What does this information tell you?

_____R3 reaches the LAN of R2 by R1. There's no direct adjacency between R2 and R3. Because S0/0/1 on R2 is passive._____

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

_____router ospf 1_____

_____no passive-interface s0/0/1_____

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

64+1 = 65

64 is serial link cost, and 1 is the R2 G0/0

LAN

Is R2 listed as an OSPF neighbor to R3? yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

Because modern links are faster than 100Mbps. By using the default reference bandwidth, links >100Mbps cost 1, OSPF cannot use faster links.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

192.168.3.0/24: R1 S0/0/1 cost + R3 G0/0 cost = 781+1=782

192.168.23.0/30: R1 S0/0/1 cost + R3 S0/0/1 = 781+781 = 1562

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562

Because each serial link has bandwidth 128kbps

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Cost = $100,000,000 / 128,000 = 781$

$781 * 2 = 1562$

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

The direct link, which is R1 to R3 is 1565, but the path through R2 is cheaper, because R1-R2 is 782, R2-R3 is 781, R3 LAN is 1, the total is $782 + 781 + 1 = 1564$. OSPF chooses the lowest accumulated cost.

~~1565~~
781 1563

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

The router id uniquely identify a router in ospf domain. If the router id changes, the whole SPF tree will be recalculated and LSAs will be flooded again.

4

2. Why is the DR/BDR election process not a concern in this lab?

Because all links between routers are p2p serial links.

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3. Why would you want to set an OSPF interface to passive?

Because setting OSPF interface to passive can stop hello packets and routing updates on it. It can improve efficiency and security.

4

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4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

Area 0



5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0



6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

EIGRP because the EIGRP's distance is 90, OSPF 's distance is 110 , RIP's distance is 120. The EIGPR have smallest administrative distances.


